

1. Administrative & Study Identification

Full Study Title: Phase I Multicenter, Open-Label, First-in-Human Study of BMS-986497 (ORM-6151) as a Monotherapy, in Double Combination With Azacitidine and in Triple Combination With Azacitidine and Venetoclax in Subjects With Relapsed or Refractory Acute Myeloid Leukemia or Myelodysplastic Syndrome
Short Title/Acronym: Phase 1 First-in-Human Study of BMS-986497 in R/R AML or MDS
Protocol Number: CA235-0001 (Other ID: 2024-519537-29)
NCT Number: NCT06419634
Posting ID: 1064103917570427088
Trial Status: RECRUITING (Currently recruiting participants)
Sponsor/Company Name: Bristol Myers Squibb
Investigational Product: BMS-986497 (ORM-6151), Azacitidine, Venetoclax
Phase of Development: Phase 1
Medical Monitor: Bristol-Myers Squibb Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective
Primary Objective: To assess the safety, tolerability, drug levels, drug efficacy and determine the recommended Phase 2 dose (RP2D) of BMS-986497 as a monotherapy, in double combination with Azacitidine, and in triple combination with Azacitidine and Venetoclax in participants with relapsed or refractory acute myeloid leukemia (AML) or myelodysplastic syndrome (MDS).
Secondary Objectives: To evaluate the pharmacokinetic profile (including Cmax, Tmax, AUC0-last), overall response rate (ORR), duration of response (DoR), best overall response (BOR), complete remission (CR) rates (including CRi and CRh), event-free survival (EFS), transition rate to allogeneic hematopoietic stem cell transplantation (HSCT), and the incidence of anti-drug antibodies (ADA) against BMS-986497.

2.2 Background & Rationale To address the critical unmet medical need in patients with relapsed or refractory AML or MDS who have failed alternative therapies with established benefit. BMS-986497 (ORM-6151) is a novel protein degrader targeting CD33 and GSPT1. This study aims to evaluate its safety and efficacy profile as a single agent and in combination regimens to potentially establish a new standard of care.

3. Study Design & Methodology

3.1 Design Framework
Study Type: Interventional
Control Type: Single-arm (Dose Escalation and Dose Expansion Cohorts)
Blinding: Open-label
Randomization: N/A (Non-randomized assignment to

monotherapy or combination groups) **Criteria Type:**
PHASE_REQUIREMENT (Trial must be in PHASE1)

3.2 Planned Enrollment & Duration Target **\$N\$:** 105 (Evaluated sequentially based on Phase 1 dose-escalation cohorts) **Number of Sites:** United States + 3 other countries (e.g., Yale University, Northwestern University, etc.) **Estimated Study Start:** 28-May-2024 **Estimated Primary Completion:** 15-Sep-2030

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults (≥ 18 years old) with primary or secondary relapsed and/or refractory acute myeloid leukemia (AML) or myelodysplastic syndrome (MDS). 2. Detectable levels of cluster of differentiation 33 (CD33) expression. 3. Failed alternative therapies with established benefit. 4. Eastern Cooperative Oncology Group (ECOG) performance status ≤ 2 and adequate organ function.

4.2 Exclusion Criteria 1. Acute Promyelocytic Leukemia. 2. Clinically active central nervous system leukemia. 3. Active malignant solid tumor. 4. Pregnant or breastfeeding. 5. Other protocol-defined exclusion criteria apply.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Administered at specified doses on specified days based on the assigned cohort (Part 1: Dose Escalation BMS-986497 Monotherapy; Part 2: Dose Expansion Combination Therapies). **Route of Administration:** * **BMS-986497:** Intravenous (IV) infusion * **Azacitidine (Preferred Name: azacitidine):** Intravenous (IV) infusion. **ATC Code:** L01BC07; **Trade Names:** Vidaza, Onureg; **EMA URLs:** [Onureg Product Info](#), [Vidaza Product Info](#). * **Venetoclax:** Oral tablet **Duration of Treatment:** Until disease progression, unacceptable toxicity, or subject withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for AML/MDS. **Prohibited:** Other investigational therapies or non-approved agents during the course of the study.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, CD33 status check, ECOG status
Baseline	Day 0	Cohort Assignment, First Dose (Monotherapy or Combination)
Interim	Per Cycle	Safety Labs, DLT Assessment, PK/PD profiling, Bone marrow biopsy for response

End of Study | Variable | Disease progression, unmanageable toxicity, or transition to HSCT |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence of dose-limiting toxicities (DLTs), incidence of treatment-emergent adverse events (TEAEs), and determination of the Recommended Phase 2 Dose (RP2D) of BMS-986497 across all monotherapy and combination arms.

Measurement Method: Clinical observation, vital signs, and laboratory assessments throughout the dose-escalation phases.

7.2 Secondary & Exploratory Endpoints * Pharmacokinetics: Cmax, Tmax, AUC0-last. * Efficacy Outcomes: Overall response rate (ORR), duration of response (DoR), best overall response (BOR), event-free survival (EFS), complete remission (CR/CRi/CRh). * Safety/Immunogenicity: Incidence of Anti-drug antibody (ADA) against BMS-986497.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring through physical examinations, vital signs, and laboratory tests for TEAEs. **Serious Adverse Events (SAEs):** Reported within 24 hours per standard oncology trial protocols. **Data Monitoring Committee (DMC):** Internal dose escalation review committee to assess DLTs before opening subsequent dose cohorts.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who receive at least one dose), DLT Evaluable Set, PK/PD Analysis Sets. **Sample Size Justification:** Target enrollment is 105 patients, guided by standard Phase 1 dose-escalation statistical models (e.g., 3+3 or Bayesian optimal interval designs) to determine MTD and RP2D safely. **Handling of Missing Data:** Handled per standard statistical practices for oncology dose-escalation trials; patients non-evaluable for DLT may be replaced.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Regular onsite and remote monitoring of sites to ensure protocol adherence and safety recording. **Audit Trail:** Thorough eCRF documentation, query resolution workflows, and data clarification forms to track clinical response and laboratory deviations.

11. Study Identifiers & Governance

Primary ID: CA235-0001 **Registry Number:** NCT06419634 **Posting ID:** 1064103917570427088 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** Study of BMS-986497 (ORM-6151) as a Monotherapy, in Double and Triple Combination With Azacitidine and Venetoclax in Participants With Relapsed or Refractory Acute Myeloid Leukemia or Myelodysplastic Syndrome **Official Regulatory Title:** Phase I Multicenter, Open-Label, First-in-Human Study of BMS-986497 (ORM-6151) as a Monotherapy, in Double Combination With Azacitidine and in Triple Combination With Azacitidine and Venetoclax in Subjects With Relapsed or Refractory Acute Myeloid Leukemia or Myelodysplastic Syndrome **Source Level:** 5

12. Clinical Framework (Disease Specific)

Primary Condition / Disease Areas: Acute Myeloid Leukemia, Myelodysplastic Syndrome **Condition Name:** Leukemia (Preferred Name: leukemia) **Preferred UMLS Name:** Myeloid Neoplasms **Semantic Type:** Neoplastic Process **Disease Definition:** A cancer of the blood and bone marrow characterized by an abnormal proliferation of leukocytes. [<https://orcid.org/0000-0002-0736-9199>, NCIT:C3161] **Diagnosis Threshold:** Pathological diagnosis of R/R AML/MDS with detectable CD33 expression **Medical Methodology:** Targeted protein degradation therapy in combination with hypomethylating agents and BCL-2 inhibitors **Optimization Target:** Identification of the Maximum Tolerated Dose (MTD) and Recommended Phase 2 Dose (RP2D) **Ontology Classifications:** * MeSH Code: D007938 * HPO Code: HP:0001909 * SNOMED ID: 93143009

13. Study Procedures & Methodology

Management Pathway: Standard clinical oncology pathway for relapsed/refractory hematological malignancies. **Education Protocol (HCP):** Site initiation visits and continuous training on DLT identification and AE management. **Education Protocol (Patient):** Comprehensive informed consent detailing phase 1 risks, administration procedures, and reporting expectations. **Quality Audit Mechanism:** Rigorous safety data review by sponsor and local IRBs prior to cohort escalation.

14. Observation & Follow-Up Schedule

Total Duration: Up to approximately 6 years (projected trial completion September 2030). **Onsite Visit Cadence:** Intensive monitoring during early cycles (e.g., Cycle 1 dose administration days) followed by regular check-ins per cycle. **Remote Monitoring Frequency:** Frequent remote review of centralized laboratory and

safety inputs. **Checklist Review Interval:** Regular review of bone marrow aspirate/biopsy results to confirm response rates and hematologic recovery profiles.

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				